

Counting Principles

Main references :

Discrete mathematics and Its Applications, 7th edition, Kenneth H. Rosen;

Discrete Mathematics with Applications, 4th edition, Susanna S. Epp

1 Basic Counting Principles

1.1 Multiplication Rule: Multiplication Principle of Sequential Counting

Theorem 1 Suppose that there are n sets of possible outcomes, denoted as $A_1, \dots, A_n$, and that the following conditions hold:

1. Choosing an outcome from one set does not affect the cardinality of the other sets, and,
2. One outcome is taken from each of the n different sets, in order from A_1 , and upto A_n .

If A is the set of all possible ways to pick the outcomes, then

$$|A| = \prod_{i=1}^n |A_i|$$

Example 1 Carvings, a fine-dining restaurant, offers a set meal. A set meal consists of one appetizer, one entrée, and one dessert. The menu has 3 dishes for appetizer, 4 dishes as entrée, and 4 dishes for dessert. How many ways can the set meal be formed?

Solution: Let A be the set of appetizer, E be the set of entrée, and D be the set of dessert. Let M be the set of all possible meal combinations from the available dishes from A , E and D . We represent a meal as a 3-tuple (a, e, d) , where $a \in A$, $e \in E$, and $d \in D$. That is, we represent the elements of M as the cross-product $A \times E \times D$. We need to count $|M|$. Choosing a dish from A does not affect the number of choices in the other sets. This is also true when we choose a dish from E and D . We can then use the multiplication rule as specified: $|M| = |A| \cdot |E| \cdot |D| = 3 \cdot 4 \cdot 4 = 48$. Hence, there are 48 possible ways to choose a meal with 3 dishes consisting of an appetizer, an entrée, and a dessert in Carvings. ■

Note that we do not always need to answer counting problems in this such detailed manner. For future solutions, we only need to identify how the choices are formed, and count them. But a word of warning, do **NOT** try to make a habit of just giving the numerical answer like:

Solution: $3 \cdot 4 \cdot 4 = 48$. ■

Try to give a bit of context to your answer in relation to the word problem. Writing bits of details for your current solution would help in how we organize our train of thought for future solutions.

Example 2 Home Security.

- (a) If a home security code consists of a sequence of two distinct english alphabet letters, how many different codes are possible? If a thief could try 5 possible codes every 30 minutes, then how long would it take the thief to try all possible codes, assuming the thief enters a different code every time?
- (b) Suppose that the home owners weren't around for a week, and when they came home, they found out they were robbed. They decided to replace the security code, which is now a four symbol code, with an alphanumeric keypad (26 letters and 10 digits).

- If repeated letters and digits are allowed in the code, how many ways to form the security code?
 - What if the code should only contain distinct letters and digits, how many ways are there to form the security code?
- (c) If a thief can input 5 codes per 30 minutes on the four symbol code security lock, how long will it take the thief to input all codes if repeated symbols are allowed? What if repeated symbols are not allowed?

Solution:

- (a) Let (c_1, c_2) , or simply c_1c_2 represent the two-letter security code. We assume that the english alphabet is the set $E = \{A, \dots, Z\}$, with 26 letters. Since the security code requires distinct letters, what was used for c_1 cannot be used for c_2 anymore. Technically, our entire choice for c_1 is E , but for c_2 , it would be E minus what was chosen for c_1 . So the actual elements of the set of letters for c_2 is affected by our choice in c_1 , but not its cardinality. There would always be 25 letters left to choose for c_2 . So, with multiplication rule, there are $26 \cdot 25 = 650$ different ways to form c_1c_2 .

Now the thief can try 5 codes per 30 minutes. Assuming that the thief does not repeat a code, it would take

$$\frac{650}{5} \cdot 30 = 130 \cdot 30 = 3900.$$

So, 3900 minutes to try out all 650 codes.

■

(So if the home owners were gone for almost 3 days, the thief could have entered their home. Assuming, the thief does not sleep, eat, nor need a bathroom break while trying out the codes.)

- (b) Let $c_1c_2c_3c_4$ represent the four-symbol alphanumeric code. There is a total of 36 alphanumeric symbols. If repetition of alphanumeric symbols is allowed, then there would be $36 \cdot 36 \cdot 36 \cdot 36 = 36^4 = 1679616$ ways to form the code. If the code has distinct symbols, then there would be $36 \cdot 35 \cdot 34 \cdot 33 = 1413720$ ways to form the four-symbol code.

■

- (c) Once again our thief comes back to try cracking the security code. We assume again that the security system only allows 5 codes per 30 minutes.

If repetition is allowed for the four-symbol alphanumeric code, then there are $36^4 = 1679616$ ways to form the code. Computing,

$$\frac{36^4}{5} \cdot 30 = 10077696.$$

So, 10077696 minutes to try out all the codes that allows repetition. That is a little over 19 years.

If the code has distinct symbols, then as previously computed, there are $36 \cdot 35 \cdot 34 \cdot 33 = 1413720$ codes in all. To try out all these codes,

$$\frac{36 \cdot 35 \cdot 34 \cdot 33}{5} \cdot 30 = 8482320$$

So, 8482320 minutes to try out all the codes with distinct symbols.

■

1.2 Addition Rule

1.2.1 Cardinality of Union of Disjoint Sets

Theorem 2 Let $A_1, \dots, A_n$ be n disjoint sets, for an integer $n \geq 1$. The cardinality of the union of the n disjoint sets is:

$$\left| \bigcup_{i=1}^n A_i \right| = |A_1| + \dots + |A_n|$$

Example 3 How many ways to get an even sum when two distinguishable 6-sided dice are rolled? What if the dice are indistinguishable?

Solution: To begin with, dice being distinguishable means that there is something (e.g. a mark, color) that can uniquely identify each die. This means that we can assign an order to the values on the dice. We let d_1 be the result/face value of the first die when rolled, and d_2 for the second die. So the result of the two dice is represented as (d_1, d_2) . Note that for d_1 and d_2 , six-sided die has possible values ranging from 1 to 6.

Now we are looking at the even sums on dice rolls, i.e. $d_1 + d_2$ is even. The even sums, and how they are formed are the following:

- Sum of 2. Formed by: $(1, 1)$.
- Sum of 4. Formed by: $(1, 3), (2, 2), (3, 1)$.
- Sum of 6. Formed by: $(1, 5), (2, 4), (3, 3), (4, 2), (5, 1)$.
- Sum of 8. Formed by: $(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)$.
- Sum of 10. Formed by: $(4, 6), (5, 5), (6, 4)$.
- and, Sum of 12. Formed by: $(6, 6)$.

Notice that no two sums share the same dice roll result. So by addition rule, we can add all dice rolls that make an even sum. Adding all the dice rolls resulting into an even sum, there are 18 ways to form the even sum with two distinguishable dice.

Now, what if the dice were indistinguishable, meaning that the two dice looks exactly alike. We cannot assign an order on the result of dice results. So a roll like $(3, 1)$ would be the same as $(1, 3)$. Instead of an ordered pair (i.e. 2-tuple), a better way to represent the result of rolling two indistinguishable dice is an unordered pair. So we represent the dice roll as $\{1, 3\}$ (looks like a set, but allows possible duplicates). Enumerating the ways to form the even sums:

- Sum of 2. Formed by: $\{1, 1\}$.
- Sum of 4. Formed by: $\{1, 3\}, \{2, 2\}$.
- Sum of 6. Formed by: $\{1, 5\}, \{2, 4\}, \{3, 3\}$.
- Sum of 8. Formed by: $\{2, 6\}, \{3, 5\}, \{4, 4\}$.
- Sum of 10. Formed by: $\{4, 6\}, \{5, 5\}$.
- and, Sum of 12. Formed by: $\{6, 6\}$.

So there is a total of 12 ways to form an even sum with two indistinguishable dice. ■

Example 4 How many 2-digit decimal numbers have their tens digit greater than the ones digit?

First, recall the Quotient-Remainder Theorem:

Theorem 3 *Quotient-Remainder Theorem.* *Let n be an integer, and d a positive integer. Then, there exists integers q , and r such that:*

$$n = dq + r, 0 \leq r < d$$

Solution: Using Theorem 3, we can represent 2-digit decimal numbers as $10d_1 + d_2$, with $1 \leq d_1 \leq 9$, and $0 \leq d_2 \leq 9$ (i.e. d_1 is the tens digit, and d_2 is the ones digit. Note as well that $d_1 > 1$ since if $d_1 = 0$, then the number is a one-digit number). So our goal is to count the possible ways of picking the value of d_1 and d_2 such that $d_1 > d_2$.

For starters, when $d_1 = 1$, then only when $d_2 = 0$ does it satisfy $d_1 > d_2$. That is the number 10. When $d_1 = 2$, then $d_2 \in \{0, 1\}$. Those are the numbers 20, and 21. Looking at the pattern, when $d_1 = i$, there are i possible values for d_2 to satisfy $d_1 > d_2$. (We do not have to do induction as you can easily list it down). We can represent the cardinality of the two-digit decimal numbers satisfying $d_1 > d_2$ as the summation:

$$\sum_{i=1}^9 i = \frac{9(9+1)}{2} = 45.$$

Hence, there are 45 two-digit decimal numbers whose tens digit is greater than its ones digit. ■

1.3 Indirect Counting

Let U and A be sets such that $A \cup \bar{A} = U$. Then, $|A| + |\bar{A}| = |U|$.

Example 5 *Consider the French deck of cards, consisting of 52 cards of 4 suites. Suppose that a card is drawn from the deck and the card is returned before the next draw (shuffling it again). In how many ways can 10 cards be drawn such that the tenth card is a repetition of a previous draw?*

Solution: Let A be the set of 10 card draws where the tenth drawn card is a repetition of any previous draws (draw 1 to draw 9). A straight-forward counting would be difficult, so we look at the complement of A , which is the set of 10 card draws where the tenth drawn card is **NOT** a repetition of any previous draws.

To count $\bar{A}$, we can first choose the card for the tenth draw (we don't have to count starting from the first draw). There are 52 possible cards to draw. Now, whichever card is chosen for the tenth, cannot be chosen for the first nine draws. But the first nine draws can be the same card. So for each of the first nine draws, there are 51 possible cards. So $|\bar{A}| = 51^9 \cdot 52$.

Now, the set of all 10 card draws without any other form of restriction or condition will be our universal set. There are 52 choices for each draw, so $|U| = 52^{10}$.

So, by indirect counting, $|A| = |U| - |\bar{A}| = 52^{10} - 51^9 \cdot 52$. ■

Note that since $52^{10} - 51^9 \cdot 52$ is a rather large number, we do not need to simplify it. For future solutions, unless the problem explicitly asks, we do not need to simplify large exponents.

Example 6 (Forbidden Arrangement) *In how many ways can 10 people be seated in a row so that a certain pair of people are not next to each other?*

Solution: Let X be the set of ways to seat 10 people so that a certain pair of people are not next to each other. We represent the 10 people to be seated as $ABCDEFGHIJ$. Rearranging the order of the letters represents the change in seating order. Now, without loss of generality, suppose that it is A and B who are not supposed to seat next to each other. So we cannot have arrangement where A and B are next together.

Now, let $\bar{X}$ be the set of ways to set 10 people where A and B **ARE** next to each other. We will treat AB as a single person, so now to count $\bar{X}$, we count the ways to arrange 9 people in order. So for the first seat, we have 9 choices. Then the second seat would have 8 choices. And so on, going down, the i th seat will have $9 - i + 1$ choices of person to seat. We can represent the cardinality of $\bar{X}$ as the product notation:

$$\prod_{i=1}^9 i = 9 \cdot 8 \cdot \dots \cdot 2 \cdot 1 = 9!$$

Now, since the arrangement AB is different from BA , then $|\bar{X}| = 2 \cdot 9!$.

Now, there are $10!$ ways to arrange 10 people to seat in a row, so $|\mathcal{U}| = 10!$. To count $|X|$, we have $|X| = 10! - 2 \cdot 9!$. ■

1.4 Inclusion-Exclusion Principle

Theorem 4 *Inclusion-Exclusion Principle (IEP)*

Let A, B, C be sets. Then,

$$|A \cup B| = |A| + |B| - |A \cap B|$$

and,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

Example 7 How many integers from 1 to 1000 are divisible by 3 or 5?

Solution: Let A_3 be the set of integers from 1 to 1000 that are divisible by 3, A_5 be the set of integers from 1 to 1000 that are divisible by 5. Need to compute $|A_3 \cup A_5|$, representing the number of integers from 1 to 1000 that are divisible by 3 or 5.

- $|A_3| = 333$. To compute this, think of which integers are divisible by 3 between 1 to 1000. We represent those integers as $3k$, for some integer k . The first integer is 3, which is $3(1)$, and the last integer is 999, which is $3(333)$. So the values of k are from 1 to 333. So there are 333 integers between 1 to 1000 that are divisible by 3.
- $|A_5| = 200$.
- $|A_3 \cap A_5| = 66$.

Hence, $|A_3 \cup A_5| = 333 + 200 - 66 = 467$. ■

Example 8 How many integers from 1 to 1000 are not divisible by 3 or 5?

Solution: From Example 7, $|A_3 \cup A_5| = 467$. Now, the set $A_3 \cup A_5$ represents the integers from 1 to 1000 that are divisible by 3 or 5. The set $\overline{A_3 \cup A_5}$ represents those integers not divisible by neither 3 nor 5. (by De Morgan's, not divisible by 3 and not divisible by 5). So we need to compute $|\overline{A_3 \cup A_5}|$. By indirect counting, $|\overline{A_3 \cup A_5}| = 1000 - 467 = 533$. ■

Example 9 The Euler phi function, or Euler's Totient, denoted as $\varphi(n)$, is the number of integers from 1 to n that are relatively prime to n . Compute $\varphi(2695)$.

Before we go into our solution, recall the definition of relatively prime:

Definition 1 Let a and b be positive integers greater than 1. a and b are said to be relatively prime when their greatest common divisor is 1. (i.e. $\text{GCD}(a, b) = 1$).

Solution: To find $\varphi(2695)$, we first need to have the prime factors of 2695. This is because if a number shares the same prime factor as 2695, then they are not relatively prime to 2695. The number 2695 has prime factorization $5 \cdot 7^2 \cdot 11$. That is, the prime factors of 2695 are 5, 7, and 11.

Let A_i be the set of integers divisible by i , for $i \in \{5, 7, 11\}$. We need to compute $|A_5 \cup A_7 \cup A_{11}|$, which represents the cardinality of the set integers from 1 to 2695 that are not divisible by any of the prime factors of 2695 (hence, relatively prime to 2695). We will use indirect counting, so first we need to find $|A_5 \cup A_7 \cup A_{11}|$. Finding the cardinalities of the individual sets,

- $|A_5| = 539$.
- $|A_7| = 385$.
- $|A_{11}| = 245$.
- $|A_5 \cap A_7| = 77$.
- $|A_5 \cap A_{11}| = 49$.
- $|A_7 \cap A_{11}| = 35$.
- $|A_5 \cap A_7 \cap A_{11}| = 7$.

By Inclusion-Exclusion Principle, we have that $|A_5 \cup A_7 \cup A_{11}| = 1015$. By Indirect Counting, with $|\mathcal{U}| = 2695$, $|A_5 \cup A_7 \cup A_{11}| = 1680$. ■

The Inclusion-Exclusion Principle can be generalized for n sets, $n \geq 2$, as follows:

Theorem 5 Generalized Inclusion-Exclusion Principle

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}| \right)$$

1.5 One-to-One Correspondence

Consider two finite sets A and B . Say that there exists a function $f: A \rightarrow B$ which is a 1-1 correspondence (i.e. f is bijective). Then $|A| = |B|$.

Suppose that we know how to count the elements of a set A , say $n = |A|$. If we manage to show that there exists a 1-1 correspondence f between A and B , then we have also counted the elements of B , that is n .

Example 10 Suppose that there are 101 players entered in a single elimination tennis tournament. In such a tournament, any player who loses a match must drop out, and every match ends in a victory for some player (that is, no ties). In each round of the tournament, the players remaining are matched into as many pairs as possible, but if there is an odd number of players left someone receives a bye (that is, an automatic victory for this player in this round). Enough rounds are played until a single player remains who wins the tournament. How many matches must be played in total?

Solution: So every match has exactly one loser. Let M be the set of matches in the tournament, and L be the set of players who lost a match. We can form a function $f : M \rightarrow L$ such that $m \in M \mapsto p \in L$ if and only if with player p losing match m . f is one-to-one correspondence since a player p who lost match m cannot continue playing anymore, so a player p can only lose exactly one match. Likewise, a match m can be defined by the player who lost match m .

Since the tournament can only end when only one player remains, it means that of the 101 players, 100 players will lose. Therefore, there will be 100 matches played in the tournament. ■

Example 11 Determine the number of subsets of a set with n elements.

Solution: Let $S = \{x_1, \dots, x_n\}$ be a set of n elements. For a subset $A \subseteq S$, for each element x_i , for all $1 \leq i \leq n$, either $x_i \in A$, or $x_i \notin A$. We represent a subset A as a binary string of length n , denoted as b_A . We define the i th bit of b_A as follows:

$$b_A(i) = \begin{cases} 1 & \text{if } x_i \in A \\ 0 & \text{if } x_i \notin A \end{cases}$$

Let X be the set of all subsets of S , and let Y be the set of all binary strings of length n . We define the function $f : X \rightarrow Y$, such that $A \in X \mapsto b_A \in Y$, with b_A defined as above. The construction of b_A for each subset A assures that each subset A is associated to exactly one binary string b_A and vice-versa. Hence, f is a one-to-one correspondence.

Now, to count $|Y|$, note that each bit of the binary string has two possibilities, a zero or a one. By multiplication rule, there are 2^n possible binary strings, that is, $|Y| = 2^n$. Since f is a one-to-one correspondence, $|X| = 2^n$. There are exactly 2^n subsets of a set with n elements. ■

Note here that the set of all subsets is referred to as the power set. For a set S , the power set of S , denoted as $\mathbb{P}(S)$, has exactly 2^n elements. Our solution to Example 11 proves this.

2 Pigeonhole Principle

Theorem 6 Pigeonhole Principle

Let X and Y be finite sets, and that $|Y| < |X|$. Then any function $f : X \rightarrow Y$ cannot be one-to-one (injective). There must be at least two elements in the domain that have the same image in the co-domain.

When doing solutions or proof that involve pigeonhole principle, first identify the set of pigeons (X), the set of pigeonholes (Y), and the function f that relates X to Y . Then if $|Y| < |X|$, we can then invoke the pigeonhole principle.

Example 12 In a group of six people, must there be at least two people who were born in the same month? What if the group consists of thirteen people?

Solution: Let X be the group of people, and Y be the set of months in a year, with $|Y| = 12$. With the function $f : X \rightarrow Y$, let $x \in X \mapsto y \in Y$ mean that person x was born on month y .

Now when there are only six people in the group, that is $|X| = 6$, then it is possible that all 6 people were born on different months. When $|X| \leq |Y|$, then it is possible to define a function f that is one-to-one. So **NO**, for a group of six people, it is not guaranteed that there will be at least two people who were born in the same month.

Now, with thirteen people in a group, that is $|X| = 13$, it is guaranteed that there will be at least two people who were born in the same month. By pigeonhole principle, with $|Y| < |X|$, there are people x_1 and x_2 that maps to the same month y . That is, these two people share the same birth month. ■

Example 13 Consider a cabinet filled white and black socks. If a sock is taken from the cabinet one at a time without looking at the color of the drawn sock, how many socks need to be taken from the cabinet to have at least one pair of socks with the same color?

Solution: Let X be the set of socks drawn from the cabinet, and Y be the colors white and black, which we denote as $\{W, B\}$, so $|Y| = 2$. For this problem, we need to find a value for $|X|$ such that $|Y| < |X|$ so we can invoke our pigeonhole principle. Now technically, any value greater than $|Y|$ will do, but what we want is the minimum. That is, $|X| = |Y| + 1 = 3$. So we need to blindly draw at least 3 socks from the cabinet to guarantee a pair of socks with the same color. ■

Illustratively, for Example 13, imagine drawing your first sock. Say it is a white sock. Then on the next draw, it is a black sock. So for the third sock drawn, its color would surely match either the first sock, or the second sock. So we only need to blindly draw three socks. (For those skeptics, we are assuming that the cabinet contains socks of the same size and design, just with colors of either white or black).

Example 14 Consider the set $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$. If five distinct integers are selected from A , must at least one pair of integers from the chosen 5 have a sum of 9?

Solution: Let $X \subseteq A$ be a selection of 5 distinct integers from A , $Y = \{\{1, 8\}, \{2, 7\}, \{3, 6\}, \{4, 5\}\}$, and a function $f : X \rightarrow Y$, such that $x \mapsto y$ if, and only if, $x \in y$. For some $x_1, x_2 \in X$, $x_1 \neq x_2$, if $f(x_1) = f(x_2)$, then $x_1 + x_2 = 9$. Since, $|X| > |Y|$, then f cannot be injective. Therefore, there is at least one pair of integers from the chosen 5 that has a sum of 9. ■

Some Elegant Applications of the Pigeonhole Principle

Example 15 During a month with 30 days, a baseball team plays at least one game a day, but no more than a total of 45 games for the whole month. Show that there must be a period of some number of consecutive days during which the team must play a total of exactly 14 games.

Proof: Let a_k be the number of games played on or before the k_{th} day of the month. Then $a_1, a_2, \dots, a_{30}$ is an increasing sequence of distinct positive integers, with $1 \leq a_k \leq 45$. Moreover, $a_1 + 14, a_2 + 14, \dots, a_{30} + 14$ is also an increasing sequence of distinct positive integers, with $15 \leq a_k + 14 \leq 59$. The 60 positive integers $a_1, a_2, \dots, a_{30}, a_1 + 14, a_2 + 14, \dots, a_{30} + 14$ are all less than or equal to 59. Hence, by the pigeonhole principle two of these integers are equal. Because the integers a_k , $k = 1, 2, \dots, 30$ are all distinct and the integers $a_k + 14$, $k = 1, 2, \dots, 30$ are all distinct, there must be indices i and j with $a_i = a_j + 14$. This means that exactly 14 games were played from day $j + 1$ to day i . ■

Example 16 Assume that in a group of six people, each pair of individuals consists of two friends or two enemies. Show that there are either three mutual friends or three mutual enemies in the group.

Proof: Let A be one of the six people. Of the five other people in the group, there are either three or more who are friends of A, or three or more who are enemies of A. This follows from the generalized pigeonhole principle, because when five objects are divided into two sets, one of the sets has at least $\left\lceil \frac{5}{2} \right\rceil = 3$ elements. In the former case, suppose that B, C, and D are friends of A. If any two of these three individuals are friends, then these two and A form a group of three mutual friends. Otherwise, B, C, and D form a set of three mutual enemies. The proof in the latter case, when there are three or more enemies of A, proceeds in a similar manner. ■

Theorem 7 Generalized Pigeonhole Principle

Let X be a set of n elements and Y be a set of m elements, with $n > m$. For any function $f : X \rightarrow Y$, there is some $y \in Y$ such that y is the image of at least $\left\lceil \frac{n}{m} \right\rceil$ distinct elements of X .

(Contrapositive form)

Let X be a set of n elements and Y be a set of m elements, with $n > m$. For any function $f : X \rightarrow Y$, if for each $y \in Y$, $f^{-1}(y)$ has at most k elements, from some positive integer k , then $n \leq km$.

We can think of the contrapositive form of GPP as setting a limit to the capacity of the pigeonhole. We limit the number of pigeons living in the same pigeonhole.

Example 17 Show that in a group of 85 people, at least 4 must have the same last name initial letter. Use both original and contrapositive form of GPP to prove this statement.

Proof: Let X be the group of 85 people, Y be the set of 26 letters of the English alphabet, and function $f : X \rightarrow Y$, such that $x \mapsto y$ if and only if, person x has last name initial y . By GPP, we have that $\frac{|X|}{|Y|} = \frac{85}{26} \approx 3.27$. Taking the ceiling, $\lceil 3.27 \rceil = 4$. Hence, at least 4 must have the same last name initial. ■

Proof: (Contrapositive GPP) Let X be the group of 85 people, Y be the set of 26 letters of the English alphabet, and function $f : X \rightarrow Y$, such that $x \mapsto y$ if and only if, person x has last name initial y . Suppose that it is not the case that there are at least 4 that have the same last name initial. That is, there at most 3 that have the same last name initial. This means that for each letter $y \in Y$, a maximum of 3 elements in X can map to this y .

By GPP, and algebra,

$$\begin{aligned} |X| &\leq 3|Y| \\ 85 &\leq 3(26) \\ 85 &\leq 78 \end{aligned}$$

which is false. What this also implies is that $|X| \leq 78$, but $|X| = 85$, so a contradiction. Hence, there are at least 4 people that have the same last name initial. ■

Example 18 What is the minimum number of cards needed to be drawn from a standard deck of 52 cards of 4 suits to guarantee that at least three cards of the same suit are chosen?

Solution: Let X : card draws, Y : 4 suits, and $f : X \rightarrow Y$, such that $x \mapsto y$ if and only if, card draw x belongs to suit y . Let $|X| = n$, and $|Y| = m = 4$. Need to find the minimum value for n such that $\left\lceil \frac{n}{m} \right\rceil = k = 3$. That is,

$$n = m(k - 1) + 1 = 4(3 - 1) + 1 = 9.$$

A minimum of 9 cards must be drawn. ■

How did we obtain that minimum $n = m(k - 1) + 1$? Recall the definition of ceiling function. $\left\lceil \frac{n}{m} \right\rceil = k$ means that,

$$\begin{aligned} k - 1 &< \frac{n}{m} \leq k \\ m(k - 1) &< n \leq km \end{aligned}$$

Since n is an integer, the minimum value of n that satisfies $m(k - 1) < n$ is $m(k - 1) + 1$.

Example 19 What is the minimum number of card draws to guarantee that the selection contains the four different suits?

Solution: Let X : card draws, Y : 4 suits, and $f : X \rightarrow Y$, such that $x \mapsto y$ if and only if, card draw x belongs to suit y . Let $|X| = n$, and $|Y| = m = 4$, and each suit $y \in Y$ has only $k = 13$ cards. Need to find the minimum value for n such that $\left\lceil \frac{n}{k} \right\rceil = m = 4$. That is,

$$n = (m - 1)k + 1 = (4 - 1)(13) + 1 = 40.$$

So we need to draw a minimum of 40 cards. ■

How is this different from Example 18? In Example 18, we are trying to minimize the number of pigeons in each hole, by using all holes, distributing all pigeons “evenly” across all holes. In Example 19, we are minimizing the number of holes used, by filling up each hole up to capacity (only works for finite capacity holes).

Example 20 There are 42 students who are to share 12 computers. Each student uses exactly one computer and no computer is used by more than 6 students. Show that at least 5 computers are used by 3 or more students.

Proof: Let X be the set of 42 students, Y the set of 12 computers, and $f : X \rightarrow Y$ a function such that, $x \mapsto y$ if, and only if, x is assigned to computer y . Additionally, each computer y can only hold 6 students at a time. Let k be the number of computers used by 3 or more students. The remaining $12 - k$ computers are then used by at most 2 students. By GPP, $42 \leq 6k + 2(12 - k)$. Solving for k , by algebra,

$$\begin{aligned} 42 &\leq 6k + 2(12 - k) \\ 42 &\leq 6k + 24 - 2k \\ 18 &\leq 4k \\ 4.5 &\leq k \end{aligned}$$

Since k is an integer, then $5 \leq k$. Therefore, at least 5 computers are used by 3 or more students. ■

3 Permutation and Combination

3.1 Permutation

Definition 2 A **permutation** is a selection or arrangement of a set of objects where ordering is important.

This is equivalent to forming an element of a cross-product of sets. We can think of permutations as an extension of our multiplication rule.

Definition 3 A permutation of k objects from a set of size n is an ordered list of k of the n objects, referred to as **k -permutation**.

Example 21 (Class Officers). At the start of a new school year, classes usually elect class officers: President, Vice-President, Secretary, and Treasurer. From a class of 30 students, if all are eligible to run for any position, and no students may be elected to more than one position, how many ways to choose the class officers?

Solution: Define P as the number of students eligible to run for class president. Similarly, VP for vice-president, S for secretary, and T for treasurer. The goal is to compute the cardinality of A , the set of ways to form the four class officers. We can choose to elect the officers in the order of P , VP , S , and then T . We have 30 choices for the P . Once the president is elected, we only have 29 choices left for VP , then 28 for S , and then finally, 27 for T . By multiplication rule, we have that $|A| = |P| \cdot |VP| \cdot |S| \cdot |T| = 30 \cdot 29 \cdot 28 \cdot 27 = 657720$. ■

Theorem 8 A permutation of k objects from a set of size n is an ordered list of k of the n objects, referred to as k -permutation. The number of k -permutations on a set of size n , denoted as $P(n, k)$, is:

$$P(n, k) = n(n-1)\dots(n-k+1) = \frac{n!}{(n-k)!}$$

Example 22 Use the k -permutation formula $P(n, k)$ for the (Class Officers) problem.

Define $n = 30$, the number of students, and $k = 4$, the number of positions to fill in a certain order. So

$$P(30, 4) = \frac{30!}{(30-4)!} = \frac{30!}{26!} = \frac{30 \cdot 29 \cdot 28 \cdot 27 \cdot 26!}{26!} = 30 \cdot 29 \cdot 28 \cdot 27$$

Corollary 1 The number of ways to put $n \geq 0$ distinct objects in order is $P(n, n) = n!$.

3.2 Combination

Definition 4 **Combination**

A **combination** is a selection or arrangement of elements of a set where ordering does not matter.

A combination of k elements from a set of size n is a subset of size k . Also referred to as **k -combination**.

Theorem 9 The number of k -combination of a set of size n , denoted as $C(n, k)$ or $\binom{n}{k}$, for $0 \leq k \leq n$, is:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Theorem 10 Let n, k be nonnegative integers, with $0 \leq k \leq n$. Then,

$$\binom{n}{k} = \binom{n}{n-k}$$

Example 23 (*Nuts or Chocolate?*) Parks and Smith, a snack-making company, has two warring groups regarding the future of the company. The 9 members of the “The League of Extraordinary Chocolates” wants to have more chocolates on the production line, while the 11 members of the “No Nuts, No Life” wants to have more bottled roasted nuts on the production line. To stop the division in the company, the president of the company decided to make a new product, a revolutionary chocolate-coated nut. To form the recipe, the president formed a committee consisting of 3 members from the chocolaty group and 4 members from the nutty group. How many ways can the president form the committee?

Solution: Let CN be the set of committee formations, C be the set of 3-member groups from the “League of Extraordinary Chocolates”, and N be the set of 4-member groups from the “No Nuts, No Life”. CN is a set of ordered pairs, with the first element from C and the second from N , so $CN = C \times N$. Hence,

$$|CN| = |C| \cdot |N| = \binom{9}{3} \cdot \binom{11}{4}$$

■

Example 24 (*Tennis Club*) Consider a tennis club of twelve members, consisting of five male and seven female. The tennis club is forming a five-person team to go to another school for practice camp.

1. How many five-person teams can be chosen that consists of three male members and two female members?
2. How many five-person teams contain at least one male member?
3. How many five-person teams contain at most one male member?

Solution:

1. Let $T_{3,2}$ be the set of five-person teams with 3 male members and 2 female members, M , the set of 3-male groups, and F , the set of 2-female groups.

$$|T_{3,2}| = \binom{5}{3} \cdot \binom{7}{2}$$

2. Let T_k be the set of five-person teams with k male members and $5-k$ female members, for $k \in \{1, \dots, 5\}$. Let $T_{k \geq 1}$ be the set of five-person teams with at least 1 male member.

$$|T_{k \geq 1}| = \binom{5}{1} \binom{7}{4} + \binom{5}{2} \binom{7}{3} + \binom{5}{3} \binom{7}{2} + \binom{5}{4} \binom{7}{1} + \binom{5}{5} \binom{7}{0}$$

3. Let T_k be the set of five-person teams with k male members and $5-k$ female members, for $k \in \{0, 1\}$. Let $T_{k \leq 1}$ be the set of five-person teams with at most 1 male member.

$$|T_{k \leq 1}| = \binom{5}{0} \binom{7}{5} + \binom{5}{1} \binom{7}{4}$$

4 Generalized Permutation and Combination

4.1 Permutation with Repetition

Theorem 11 *The number of r -permutations of a set of n objects with repetition allowed is n^r .*

Example 25 *How many strings of length r can be formed using the alphabet:*

1. $\{0, 1\}$. There are 2^r binary strings.
2. $\{0, \dots, 7\}$. There are 8^r octal strings.
3. $\{0, \dots, 9, A, \dots, F\}$. There are 16^r hexadecimal strings.

Permutation with Indistinguishable Objects

Example 26 (*MISSISSIPPI with One Eye*). *How many ways can we form a word by rearranging the letters of the string “MISSISSIPPI”?*

Solution: Think of the indices of the 11 characters of the string as a set of integers from 1 to 11, $A = \{1, \dots, 11\}$. If we want to place the character I in four of the 11 available positions, then we can think of it as obtaining a 4-element subset of A . Taking the subset $\{1, 4, 7, 11\}$ would mean putting the letter I on indices 1, 4, 7, and 11. The number of ways to choose the 4-sized subset is $\binom{11}{4}$. After choosing a position for I , there will be 7 positions left to fill.

Doing a similar process, we have $\binom{7}{4}$ ways to choose the indices for character S , $\binom{3}{2}$ for character P and $\binom{1}{1}$ for character M . The multiplication rule can then be used to obtain the number of ways the characters I , S , P , and M can be arranged. That is,

$$\binom{11}{4} \binom{7}{4} \binom{3}{2} \binom{1}{1} = \frac{11!}{4!4!2!1!} = \binom{11}{4, 4, 2, 1}$$

Note here that the ordering of which character is assigned a set of indices first does not matter. Do take note the number of remaining available indices for the characters we have yet to place. Example, if we choose to assign the indices for characters M , I , P , and S , in order:

$$\binom{11}{1} \binom{10}{4} \binom{6}{2} \binom{4}{4} = \frac{11!}{1!4!2!4!} = \binom{11}{4, 4, 2, 1}$$

■

Note the notation $\binom{11}{4, 4, 2, 1}$ used in Example 26 is referred to as a multinomial (the combination is alternatively referred to as a binomial).

Theorem 12 *The number of different permutations of k distinct objects, with multiplicity of each distinct object represented as $m_1, m_2, \dots, m_k$ is*

$$\frac{(m_1 + m_2 + \dots + m_k)!}{m_1!m_2!\dots m_k!}$$

4.2 Combination with Repetition

Definition 5 An r -combination with repetition allowed, or **multiset of size r** , chosen from a set X of n elements is an unordered selection of elements taken from X with repetition allowed. If $X = \{x_1, \dots, x_n\}$, we write an r -combination with repetition allowed, or multiset of size r , as $[x_{i_1}, x_{i_2}, \dots, x_{i_r}]$ where each x_{i_j} is in X and some of the x_{i_j} may equal each other.

Example 27 Let $X = \{a, b, c\}$. Then some of the possible multisets over X are:

- $[a, b, b, c, a]$
- $[a, a, a, a]$
- $[\]$. This is referred to as the empty multiset.

Combination without repetition can be thought of as counting the number of subsets a finite set has. Combination with repetition can be thought of as counting how many finite sum multisets can be formed from the finite set. Some texts refer to this as r -multicombination of n elements. We will refer to r -combination with repetition as either a multiset of size r , or r -multicombination, from now on.

Example 28 How many multisets of three elements can be made from the set $\{1, 2, 3, 4\}$?

Solution: Consider the multisets of size 3 that we can form from the set $X = \{1, 2, 3, 4\}$. We can use an exhaustive counting, listing down all possible multisets of size 3 to count them all. This approach is cumbersome. What if we wanted a bigger multiset, or if the set we are forming the multisets from has more elements? We need a different approach.

Consider two multisets from X : $[1, 1, 2]$, and $[1, 3, 4]$. We can represent each multiset as box with dividers. Think of a long box, and place three dividers inside the box. It now has four compartments. Each compartment will represent an element of X . The box,

xx	x		
----	---	--	--

represents $[1, 1, 2]$ as there are two objects in the first compartment, and one object in the second compartment. The box,

x		x	x
---	--	---	---

represents $[1, 3, 4]$ as there is one object each in compartment one, three, and four. Now let us abstract the box, and represent it as a string of $|$ and x . So $[1, 1, 2]$ is represented as $xx|x|$, while $[1, 3, 4]$ is represented as $x||x|x$.

Each arrangement of three $|$ and three x will represent a unique multiset of size 3 formed from the set X . Hence, the problem now is how many ways can we arrange three $|$ and three x . This is a problem similar to permutation with indistinguishable objects. By Theorem 12, the number of arrangements of three $|$ and three x would be,

$$\frac{(3+3)!}{3!3!} = \frac{6!}{3!3!} = \binom{6}{3}$$

■

We can summarize the result of our solution in Example 28 as the following theorem:

Theorem 13 There are $\binom{r+n-1}{r}$ r -multicombinations from a set with n elements when repetition of elements is allowed.

Remember that r is the size of the multiset, and n is the size of set we form the multiset from. A word of caution, a common mistake is to have $r - 1$ in the denominator of the combination, instead of the correct r , or having n instead of the correct $n - 1$.

We will use Theorem 13 for the following example.

Example 29 *A player has to draw five tickets from a raffle box. The raffle box contains 4 different tickets: platinum tickets, gold tickets, silver tickets, and bronze tickets, each representing a specified cash prize. Assuming that there are at least five for each kind of ticket, how many ways are there to draw the five tickets?*

Solution: Now we must identify the components of the formula from Theorem 13. We need to draw five tickets from a box containing 4 different kinds of tickets. We represent the set of tickets as $T = \{P, G, S, B\}$, representing the different ticket values. A draw of five tickets is a multiset of size 5. So we are doing 5-multicombination of 4 elements. By Theorem 13, we have that

$$\binom{5+4-1}{5} = \binom{8}{5} = \frac{8 \cdot 7 \cdot 6 \cdot 5!}{3! \cdot 5!} = 56$$

So there are 56 ways to draw the five tickets from a box of four possible ticket prizes. ■

Note here that Example 29 explicitly states that there are at least five for each kind of ticket. This is because when we form the multiset, we can assume that we can fill it up with just a single kind of ticket. But realistically, no contest would put in as much grand prize ticket as a common prize ticket. If the platinum ticket is the grand prize, there is usually just one of it inside the box.

Our next example demonstrates the problem of having a certain element with limited multiplicity.

Example 30 *You were asked to run to the nearest convenience store to buy fifteen cans of soda. The convenience store has five different brands of soda to choose from.*

1. *Assuming that there are at least fifteen cans of each brand of soda, how many ways can the fifteen cans be selected?*
2. *If 7-Sprite is one of the brands, how many selections of the fifteen cans include at least six cans of 7-Sprite?*
3. *If the store has only five cans of 7-Sprite left but at least fifteen cans of the others, how many different selections are there?*

Solution:

1. Let $n = 5$, the number of brands of soda, and $r = 15$, the number of cans of soda to buy. The selection can contain cans with the same brand of soda. So,

$$\binom{15+5-1}{15} = \binom{19}{15}$$

2. Since 6 of the 15 cans are already fixed to be 7-Sprite soda, we only need to choose for the 9 remaining cans. Note that we can still choose 7-Sprite cans.

$$\binom{9+5-1}{9} = \binom{13}{9}$$

3. Let A_i be the set of selections of 15 soda cans with i 7-Sprite cans. Since there are only 5 7-Sprite cans remaining, $i \in \{0, \dots, 5\}$. We compute for $\left| \bigcup_{i=0}^5 A_i \right|$. Note that these selections are mutually disjoint, so the addition rule can be used.

$$\begin{aligned} \left| \bigcup_{i=0}^5 A_i \right| &= \binom{15+4-1}{15} + \binom{14+4-1}{14} + \binom{13+4-1}{13} + \binom{12+4-1}{12} \\ &\quad + \binom{11+4-1}{11} + \binom{10+4-1}{10} \\ &= \binom{18}{15} + \binom{17}{14} + \binom{16}{13} + \binom{15}{12} + \binom{14}{11} + \binom{13}{10} \end{aligned}$$

Alternatively, using Indirect counting, $\left| \bigcup_{i=0}^5 A_i \right|$ can be viewed as a single set $A_{i \leq 5}$, the selections with at most 5 7-Sprite cans, and its complement, $A_{i \geq 6}$, the selections with at least 6 7-Sprite cans. So,

$$|A_{i \leq 5}| = |\mathcal{U}| - |A_{i \geq 6}| = \binom{19}{15} - \binom{13}{9}$$

4.3 Distinguishable Objects and Distinguishable Boxes

Theorem 14 *The number of ways to distribute n distinguishable objects into k distinguishable boxes so that n_i objects are placed into box i , for $i = 1, 2, \dots, k$ is*

$$\frac{n!}{n_1! n_2! \cdots n_k!}$$

We can observe that this problem is similar to permutation with indistinguishable objects.

Example 31 *Consider the standard deck of 52 cards of 4 suits. How many ways are there to distribute hands of 13 cards to each of four players?*

Solution: Let the four players be A , B , C , and D . There are $\binom{52}{13}$ ways for A to get the 13 cards from the still undistributed deck. Then for the remaining 39 cards, there are $\binom{39}{13}$ ways for B to get 13 cards. Then $\binom{26}{13}$ ways for C to get 13 cards, and finally $\binom{13}{13}$ ways for D to get the final 13 cards. So,

$$\binom{52}{13} \binom{39}{13} \binom{26}{13} \binom{13}{13} = \frac{52!}{13!13!13!13!}$$

as indicated by Theorem 14. ■

Now if we do not care which player gets each set of 13 cards per distribution, that is, we do not care about ordering of the players, then we can further divide the result by $4!$.

Example 32 *Consider the standard deck of 52 cards. How many ways are there to distribute hands of 5 cards to each of four players?*

Solution: This example is a bit tricky. The solution is similar to Example 31, but this time, there will be some cards remaining in the deck after distribution to the four players. We also have to take into consideration the cards that remain in the deck. Think of the deck as a 5th player who gets all the remaining undistributed cards. By Theorem 14, there are

$$\frac{52!}{5!5!5!32!}$$

ways to distribute 5 cards to each of the four players. ■

4.4 Indistinguishable Objects and Distinguishable Boxes

Example 33 *How many ways are there to place 10 indistinguishable balls into 8 distinguishable bins?*

Solution: Similar to Example 28, we can consider the 8 distinguishable bins as a single long box, with 7 dividers, where we can label each of the 8 compartments to correspond to each bin. We assume that each compartment can hold at least 10 balls. By representing the arrangement of placing 10 balls into 8 bins as a string of $\circ$ and $|$, the problem is now counting the ways to form strings with 10 $\circ$ and 7 $|$. Essentially, the problem is now finding the number of 10-multicombinations of a set with 8 elements. By Theorem 13, we have $\binom{10+8-1}{10}$ ways to place 10 indistinguishable balls into 8 distinguishable bins. ■

Similar to r -multicombinations of n elements, the number of ways to put in r indistinguishable objects into n distinguishable boxes is

$$\binom{r+n-1}{r} = \binom{r+n-1}{n-1}$$

Example 34 *Consider the equation:*

$$x_1 + x_2 + x_3 = 5$$

How many integer solutions are there when for each $i = 1, 2, 3$, $x_i \geq 0$?

Solution: An example integer solution to the equation is $x_1 = 2, x_2 = 1, x_3 = 2$. We can represent this as a multiset $[x_1, x_1, x_2, x_3, x_3]$. We are placing 5 indistinguishable objects into 3 bins, each labeled with x_1, x_2 , and x_3 . So we are counting the number of 5-multicombinations of 3 elements. So there are $\binom{5+3-1}{5}$ integer solutions to the equation. ■

As a follow-up, consider the problem when we have each $x_i \leq 3$, for $1 \leq i \leq 3$. We will have to use a bit of Inclusion-Exclusion Principle with a dash of Indirect Counting to solve this problem with additional conditions on each variable x_i .

4.5 Distinguishable Objects and Indistinguishable Boxes

Theorem 15 *Stirling number of the second kind*

The Stirling number of the second kind is the number of ways to partition a set of n elements into j non-empty subsets. It is denoted as $S(n, j)$ or $\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}$.

$$\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} = \frac{1}{j!} \sum_{i=0}^{j-1} (-1)^i \binom{j}{i} (j-i)^n$$

Note: if the indistinguishable boxes can be empty, then the number of ways to distribute n distinguishable objects into k indistinguishable boxes is

$$\sum_{j=1}^k \left\{ \begin{matrix} n \\ j \end{matrix} \right\}$$

Example 35 How many ways are there to put four different employees into three indistinguishable offices, when each office can contain any number of employees?

Solution: We will use Theorem 15 to solve this problem. Since there are four employees to distribute, $n = 4$. So we consider if the four employees are distributed into: (1) only one room, (2) two rooms, and (3) all three rooms.

$$\left\{ \begin{matrix} 4 \\ 1 \end{matrix} \right\} + \left\{ \begin{matrix} 4 \\ 2 \end{matrix} \right\} + \left\{ \begin{matrix} 4 \\ 3 \end{matrix} \right\} = 1 + 7 + 6 = 14$$

$$\begin{aligned} \left\{ \begin{matrix} 4 \\ 1 \end{matrix} \right\} &= \binom{1}{0} (1-0)^4 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \left\{ \begin{matrix} 4 \\ 2 \end{matrix} \right\} &= \frac{1}{2} \left(\binom{2}{0} (2-0)^4 - \binom{2}{1} (2-1)^4 \right) \\ &= \frac{1}{2} (16 - 2) \\ &= 7 \end{aligned}$$

$$\begin{aligned} \left\{ \begin{matrix} 4 \\ 3 \end{matrix} \right\} &= \frac{1}{6} \left(\binom{3}{0} (3-0)^4 - \binom{3}{1} (3-1)^4 + \binom{3}{2} (3-2)^4 \right) \\ &= \frac{1}{6} (81 - 3 \cdot 16 + 3) \\ &= 6 \end{aligned}$$

■

4.6 Indistinguishable Objects and Indistinguishable Boxes

Example 36 How many ways are there to pack six copies of the same books into four identical boxes, where a box can contain as many as six books?

Observe that distributing n indistinguishable objects into k indistinguishable boxes is the same as writing n as the sum of at most k positive integers in decreasing order.

If $p_k(n)$ is the number of partitions of n into at most k positive integers, then there are $p_k(n)$ ways to distribute n indistinguishable objects into k indistinguishable boxes.

No simple closed formula exists for this number.

5 Combinatorial Proof, Pascal's Formula and Binomial Theorem

In this section, we introduce the notion of a **combinatorial proof**. A **combinatorial proof** is a method of proof that uses counting principles in order to prove the truth of a statement. In contrast, we refer to a proof that focuses on the use of algebraic rules as an **algebraic proof**.

A number of theorems can be proven by both combinatorial proofs and algebraic proofs. We demonstrate this with Example 37 and Theorem 16.

Example 37 *Prove the equality:*

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

Proof: (By algebra)

$$\begin{aligned} k \binom{n}{k} &= k \cdot \frac{n!}{k!(n-k)!} \\ &= \frac{k \cdot n \cdot (n-1)!}{k \cdot (k-1)!(n-k)!} \\ &= n \cdot \frac{(n-1)!}{(k-1)!(n-k-1+1)!} \\ &= n \cdot \frac{(n-1)!}{(k-1)!((n-1)-(k-1))!} \\ &= n \binom{n-1}{k-1} \end{aligned}$$

■

Proof: (Combinatorial Proof) We can think of the problem as forming a committee of k members with a leader from a group of n people. There are two ways to do this.

(First method) First, we choose k members of the committee from n people. There are $\binom{n}{k}$ ways to do this. Then, from the k members, we choose one leader. Assuming every member is eligible to be the leader, then there are k choices for the leader. Hence, the number of ways to form the committee with a leader is $k \binom{n}{k}$.

(Second method) First, we choose the leader of the committee from the group of n people. Assuming every person is eligible, then there are n choices for the leader. After choosing the leader, we only need to choose $k-1$ members of the committee from the remaining $n-1$ people. There are $\binom{n-1}{k-1}$ to form the remaining members of the committee. Hence, the number of ways to form the committee with a leader is $n \binom{n-1}{k-1}$.

Both methods describe how to form the committee of k members with a leader from a group of n people. Hence,

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

■

Theorem 16 *Pascal's Formula*

Let $n, r \in \mathbb{Z}^+$, with $n \geq r$. Then,

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$$

Proof: (By algebra)

$$\begin{aligned}
\binom{n}{r} + \binom{n}{r-1} &= \frac{n!}{r!(n-r)!} + \frac{n!}{(r-1)!(n-r+1)!} \\
&= \frac{(n+1-r)n! + rn!}{r!(n+1-r)!} \\
&= \frac{(n+1)n!}{r!(n+1-r)!} \\
&= \frac{(n+1)!}{r!(n+1-r)!} \\
&= \binom{n+1}{r}
\end{aligned}$$

■

Proof: (Combinatorial Proof) Let r, n be positive integers with $r \leq n$. Let S be a set of $n+1$ elements, $S = \{x_1, \dots, x_{n+1}\}$. The set of r -element subsets of r can be partitioned into two: (A) subsets not containing x_{n+1} and (B) subsets containing x_{n+1} . Observe that A is a set complement to B .

Subsets in A contain r elements from the set $S - \{x_{n+1}\}$, while subsets in B contain $r-1$ elements from the set $S - \{x_{n+1}\}$. Note that $S - \{x_{n+1}\}$ has n elements. Hence, $|A| = \binom{n}{r}$, and $|B| = \binom{n}{r-1}$. By addition rule,

$$\begin{aligned}
|S| &= |A| + |B| \\
\binom{n+1}{r} &= \binom{n}{r} + \binom{n}{r-1}
\end{aligned}$$

as needed to be shown.

■

The next theorem can also be proven by both algebraic proof and combinatorial proof. But the algebraic proof is long and involves a lot of reindexing of summations. As such, the combinatorial proof is much more appealing in this cases.

Theorem 17 Binomial Theorem

$\forall x, y \in X$, where $X \in \{\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{C}\}$, and, $\forall n \in \mathbb{N}$,

$$(x+y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$$

Proof: (Combinatorial Proof) Let a and b be real numbers and $n \geq 1$ be a positive integer. The expression $(a+b)^n$ can be expanded as sums of products of n letters (a's and b's). For each $k \in \{0, \dots, n\}$, the number of occurrences of the product $a^{n-k}b^k$ as a term in the sum is the same as the number of ways of ordering $(n-k)$ a's and k b's. That is, $\frac{n-k+k}{(n-k)!k!} = \binom{n}{k}$.

Combining like terms, the coefficient of the term $a^{n-k}b^k$ is $\binom{n}{k}$. Therefore,

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

as needed to be shown.

For $n = 0$, $(a + b)^0 = 1$, and that

$$\sum_{k=0}^0 \binom{n}{k} a^{n-k} b^k = \binom{0}{0} a^0 b^0 = 1$$

as needed to be shown. ■

Theorem 16 and Theorem 17 can be used to determine the coefficients of the expansion of $(x + y)^n$, for $n \geq 0$. The listing of the coefficient is represented using Pascal's Triangle in Figures 1 and 2.

$$\begin{array}{c} \binom{0}{0} \\ \binom{1}{0} \quad \binom{1}{1} \\ \binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2} \\ \binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3} \\ \binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4} \end{array}$$

Figure 1: Pascal's Triangle for the binomial coefficient of $(x + y)^n$ for $n = 0, 1, 2, 3, 4$.

$$\begin{array}{c} 1 \\ 1 \quad 1 \\ 1 \quad 2 \quad 1 \\ 1 \quad 3 \quad 3 \quad 1 \\ 1 \quad 4 \quad 6 \quad 4 \quad 1 \end{array}$$

Figure 2: Pascal's Triangle for the evaluated coefficient of $(x + y)^n$ for $n = 0, 1, 2, 3, 4$.

Example 38 Compute the coefficients of the following terms:

1. 18th term of $(x + 2y)^{25}$.

Solution: $\binom{25}{18} x^7 (2y)^{18}$. Note that the term with $k = 0$ is the zeroth term.

2. 10th term of $(2x - 3y)^{15}$

Solution: $\binom{15}{10} (2x)^5 (3y)^{10}$.

6 Generating Permutations

Sometimes, permutations or combinations need to be generated, not counted.

Consider the following:

- Suppose that a salesman must visit six different cities. In which order should these cities be visited to minimize total travel time?
- Suppose we are given a set of six positive integers and wish to find a subset of them that has 100 as their sum, if such a subset exists.
- Suppose a laboratory has 95 employees where each employee can have one or more skills. A group of 12 of these employees with a particular set of 25 skills is needed for a project.

Lexicographical Ordering

In this ordering, the permutation $a_1a_2 \cdots a_n$ precedes the permutation $b_1b_2 \cdots b_n$, if for some k , with $1 \leq k \leq n$, $a_1 = b_1, a_2 = b_2, \dots, a_{k-1} = b_{k-1}$, and $a_k < b_k$.

(i.e. the permutation with the smaller number in the first position where the permutations disagree goes first)

Example 39 Permutation of 1, 2, 3, 4, 5

- 12345 is the “smallest” permutation
- 54321 is the “largest” permutation
- 23415 precedes 23514

Algorithm 1: Generating the Next Permutation in Lexicographical Order

Input: $a_1a_2 \cdots a_n$: permutation of $\{1, 2, \dots, n\}$ not equal to $n \ n-1 \ \dots \ 2 \ 1$

Output: $a'_1a'_2 \cdots a'_n$: permutations that succeeds $a_1a_2 \cdots a_n$ in lexicographical order.

$j \leftarrow n - 1$

while $a_j > a_{j+1}$ **do**

$j \leftarrow j - 1$

// j is the largest subscript with $a_j < a_{j+1}$

$k \leftarrow n$

while $a_j > a_k$ **do**

$k \leftarrow k - 1$

// a_k is the smallest integer greater than a_j to the right of a_j

interchange a_j and a_k

$r \leftarrow n$

$s \leftarrow j + 1$

while $r > s$ **do**

 interchange a_r and a_s

$r \leftarrow r - 1$

$s \leftarrow s + 1$

// this puts the tail end of the permutation after the j th position in increasing order

return $a_1a_2 \cdots a_n$

// $a_1a_2 \cdots a_n$ is now the next permutation

7 Generating Combinations

A subset of the set $\{1, 2, \dots, n\}$ (or any n -element set) can be represented by a binary string of length n . Consider the set $\{1, 2, 3, 4\}$. The subset, $\{1, 2, 3\}$ and $\{2, 4\}$ is represented by 1110 and 0101, respectively. Hence, for any subset H of $S = \{a_1, a_2, \dots, a_n\}$, the binary string $s_H = s_1 s_2 \dots s_n$ that corresponds to H is defined as:

$$s_i = \begin{cases} 1 & a_i \in H \\ 0 & a_i \notin H \end{cases}$$

for all i , $1 \leq i \leq n$.

To enumerate all subsets of an n -element set, start with the subset corresponding to 000...00. Obtain the next subset by performing binary addition, adding 1 to the binary string of the previous subset. Do this until the subset corresponding to 111...11 is obtained.

Generating r -combinations

An r -combination can be represented by a sequence containing the elements in the subset in increasing order. The r -combination can be listed using lexicographical order on these sequences. In lexicographical ordering, the first r -combination is $\{1, 2, \dots, r-1, r\}$ and the last r -combination is $\{n-r+1, n-r+2, \dots, n-1, n\}$. Given an r -combination $\{a_1, \dots, a_r\}$, the next r -combination can be obtained by **Algorithm 2**.

Algorithm 2: Generating the Next r -Combination in Lexicographical Order

Input: $\{a_1, a_2, \dots, a_r\}$: proper subset of $\{1, 2, \dots, n\}$ not equal to $\{n-r+1, \dots, n\}$
with $a_1 < a_2 < \dots < a_r$

Output: $\{a'_1, a'_2, \dots, a'_r\}$: r -element subset that succeeds $\{a_1, a_2, \dots, a_r\}$ in
Lexicographical Order

$i \leftarrow r$;

while $a_i = n - r + i$ **do**

$i \leftarrow i - 1$;

$a_i \leftarrow a_i + 1$;

for $j \leftarrow i + 1$ **to** r **do**

$a_j \leftarrow a_i + j - i$;

return $\{a_1, a_2, \dots, a_r\}$;

8 Recurrence Relations

Definition 6 A **recurrence relation** for a sequence $a_0, a_1, a_2, \dots$ is a formula that relates each term a_k to certain of its predecessors $a_{k-1}, a_{k-2}, \dots, a_{k-i}$, where i is an integer with $k-i \geq 0$. The initial conditions for such a recurrence relation specify the values of $a_0, a_1, a_2, \dots, a_{i-1}$, if i is a fixed integer, or $a_0, a_1, \dots, a_m$, where m is an integer with $m \geq 0$, if i depends on k .

Example 40

- Let $h_0, h_1, h_2, \dots$ be defined as:

$$\begin{aligned} h_0 &= 1, h_1 = 2, h_2 = 3 \\ \forall n \geq 3, h_n &= h_{n-1} + h_{n-2} + h_{n-3} \end{aligned}$$

- Let f be defined recursively as $f(0) = 3$, $f(n+1) = 2f(n) + 3$ for $n \geq 1$.

A sequence, $\{a_n\}$, is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

A recurrence relation can be used to model solutions to counting problems.

Example 41 Fibonacci. *A young pair of rabbits (one of each sex) is placed on an island. A pair of rabbits does not breed until they are 2 months old. After they are 2 months old, each pair of rabbits produces another pair each month. Find a recurrence relation for the number of pairs of rabbits on the island after n months, assuming that no rabbits ever die.*

Solution: Let f_n denote the number of rabbit pairs after n months. Show that $\{f_n\}_{n \geq 1}$ are the terms of the Fibonacci sequence. $f_1 = 1$, $f_2 = 1$, and for all $n \geq 3$, $f_n = f_{n-1} + f_{n-2}$.

Example 42 Tower of Hanoi.

(A popular puzzle invented in the late nineteenth century by Édouard Lucas.)

In the Towers of Hanoi problem there are three pegs (posts) and n disks of different sizes. Each disk has a hole in the middle so that it can fit on any peg. At the beginning of the game, all n disks are on the first peg, arranged such that the largest is on the bottom, and the smallest is on the top (thus the first peg looks like a tower).

The goal of the game is to end up with all disks on another peg, in the same order, that is, smallest on top, and increasing size towards the bottom.

But, there are some restrictions to how the disks are moved (which make the problem non-trivial):

1. *The only allowed type of move is to grab one disk from the top of one peg and drop it on another peg. That is, you cannot grab several disks at one time.*
2. *A larger disk can never lie above a smaller disk, on any post.*

Let H_n denote the number of moves needed to solve the Tower of Hanoi with n disks. Find a recurrence relation for the sequence $\{H_n\}_{n \geq 1}$.

Solution:

To transfer one disk from one peg to another, we only need one move. So $H_1 = 1$.

Now, consider a tower of $n \geq 2$ disks. We label the three pegs as: the initial peg where the disks are initially positioned, the destination peg where we transfer all the disks to, and the extra peg which we use for extra space.

To transfer the largest disk (the one at the bottom of the tower) to the destination peg, we have to first move the $n - 1$ disks on top of it to the extra peg. We need H_{n-1} moves to do this. Now, once free, we can then transfer the largest disk on the destination peg, taking one move. And then finally, we transfer the $n - 1$ disks from the extra peg to the destination peg, on top of the largest disk, which will take another H_{n-1} moves. So, to move n disks, $H_n = 2H_{n-1} + 1$.

$$H_1 = 1; \forall n \geq 2, H_n = 2H_{n-1} + 1$$

Alternatively, we propose that H_n has a closed form. That is,

$$H_n = 2^n - 1$$

Proof: (By Induction)

Basis Step: $H_1 = 2^1 - 1 = 2 - 1 = 1$. So our closed formula gives the 1st term that satisfies the recurrence relation.

Inductive Step: Suppose that for $n \geq 1$, that $H_i = 2^i - 1$, for all i , $1 \leq i \leq n$. . Need to show that $H_{n+1} = 2^{n+1} - 1$. By algebra,

$$\begin{aligned} H_{n+1} &= 2H_n + 1 \\ &= 2(2^n - 1) + 1 \\ &= 2^{n+1} - 2 + 1 \\ &= 2^{n+1} - 1 \end{aligned}$$

■

Example 43 Find a recurrence relation and give initial conditions for the number of bit strings of length n that do not have two consecutive 0s. How many such bit strings are of length five?

Solution: Let a_n denote the number of bit strings of length n that do not have two consecutive 0s. By addition rule, a_n is the number of bit strings of length n ending with a 0 with no two consecutive 0s plus those that end with a 1 with no two consecutive 0s.

The bit strings of length n that ends with 1 with no two consecutive 0s is equal to the number of bit strings of length $n - 1$ with no two consecutive 0s concatenated with a 1 at the end. That is, there are a_{n-1} such bit strings.

The bit strings of length n that ends with 0 with no two consecutive 0s must have a bit 1 on the $(n - 1)$ st bit. Hence, the number of such strings is the number of bit strings of length $n - 2$ that contains no two consecutive 0s with 10 concatenated at the end. There are a_{n-2} such bit strings.

The recurrence relation is: $a_n = a_{n-1} + a_{n-2}$, with the initial condition that $a_1 = 2$ and $a_2 = 3$.

Note that this has the same recurrence relation as the Fibonacci sequence. $a_1 = f_3$, $a_2 = f_4$ and so $a_n = f_{n+2}$, for all $n \geq 3$.

■

Example 44 Codeword Enumeration. A computer system considers a string of decimal digits a valid codeword if it contains an even number of 0 digits. (e.g. 1230407869 is valid, while 120987045608 is not valid.) Let a_n be the number of valid n -digit codewords. Find a recurrence relation for a_n .

Solution: As a base case, there are 9 valid 1-digit code word (1...9) from the 10 possible 1-digit code word (0 is not a valid codeword). For the n -digit code word, there are two ways to form a valid n -digit codeword from a $n - 1$ -digit codeword.

1. For all valid $(n - 1)$ -digit codeword, append any digit from 1 to 9. There would be $9a_{n-1}$ of this new valid n -digit codeword.
2. For all invalid $(n - 1)$ -digit codeword, append digit 0 to make a valid n -digit codeword. By indirect counting, there would be $10^{n-1} - a_{n-1}$ invalid $(n - 1)$ -digit codeword, hence forming the same number of new n -digit codeword with digit 0 at the end.

By addition rule,

$$\begin{aligned} a_n &= 9a_{n-1} + (10^{n-1} - a_{n-1}) \\ &= 8a_{n-1} + 10^{n-1} \end{aligned}$$

with $a_1 = 9$.

■

Example 45 Find a recurrence relation for C_n , the number of ways to parenthesize the product of $n + 1$ numbers, $x_0 \cdot x_1 \cdot x_2 \cdot \dots \cdot x_n$, to specify the order of multiplication.

Solution:

The parentheses can be inserted in the product $x_0 \cdot x_1 \cdot \dots \cdot x_n$, with one multiplication that is outside all parentheses (i.e. the final multiplication operation). The variables $x_0, x_1, \dots, x_n$, can be partitioned into two, splitting at x_k and x_{k+1} . That is, find the number of ways to order the multiplication of $x_0 \cdot x_1 \cdot \dots \cdot x_k$ multiplied with the number of ways to order the multiplication of $x_{k+1} \cdot x_{k+2} \cdot \dots \cdot x_n$. That is, $C_k C_{n-k-1}$. Do this for all k , $0 \leq k \leq n-1$. By addition rule,

$$\begin{aligned} C_n &= C_0 C_{n-1} + C_1 C_{n-2} + \dots + C_{n-2} C_1 + C_{n-1} C_0 \\ &= \sum_{k=0}^{n-1} C_k C_{n-k-1} \end{aligned}$$

where $C_0 = 1$ and $C_1 = 1$. ■

The sequence $\{C_n\}$ is the sequence of *Catalan numbers* named after Eugène Charles Catalan.

9 Solving Linear Recurrence Relations

Definition 7 A linear homogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

where $c_1, \dots, c_k$ are real numbers, and $c_k \neq 0$.

Note that a degree k recurrence relation requires k initial conditions.

Example 46

1. $P_n = (1.11)P_{n-1}$ is a linear homogeneous recurrence relation of degree one.
2. $f_n = f_{n-1} + f_{n-2}$ is a linear homogeneous recurrence relation of degree two.
3. $a_n = a_{n-5}$ is a linear homogeneous recurrence relation of degree five.
4. $a_n = a_{n-1} + a_{n-2}^2$ is not linear.
5. $H_n = 2H_{n-1} + 1$ is not homogeneous.
6. $B_n = nB_{n-1}$ does not have constant coefficients.

9.1 Solving Linear Homogeneous Recurrence Relations with Constant Coefficients

The basic approach is to look for solutions of the form $a_n = r^n$, where r is a constant. $a_n = r^n$ is a solution to $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ if, and only if,

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

Dividing by r^{n-k} , and subtracting the right-hand side from both sides,

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0 \quad (1)$$

Equation 1 is referred to as the characteristic equation of the recurrence relation. The solutions to this equation are referred to as the characteristic roots of the recurrence relation.

Theorem 18 Let c_1 and c_2 be real numbers. Suppose that $r^2 - c_1r - c_2 = 0$ has two distinct roots, r_1 and r_2 . Then the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1r_1^n + \alpha_2r_2^n$, for all $n \geq 0$, where α_1 and α_2 are constants.

Given initial values a_0 and a_1 , we can solve for the constants α_1 and α_2 . To solve for α_1 and α_2 , assuming that $r_1 \neq r_2$, note that,

$$\begin{aligned} a_0 &= C_0 = \alpha_1 + \alpha_2 \\ a_1 &= C_1 = \alpha_1r_1 + \alpha_2r_2 \end{aligned}$$

Solving this system of linear equations,

$$\begin{aligned} C_1 &= \alpha_1r_1 + (C_0 - \alpha_1)r_2 \\ &= \alpha_1(r_1 - r_2) + C_0r_2 \end{aligned}$$

Hence,

$$\alpha_1 = \frac{C_1 - C_0r_2}{r_1 - r_2}$$

and

$$\alpha_2 = C_0 - \alpha_1 = \frac{C_0r_1 - C_1}{r_1 - r_2}$$

Example 47 What is the solution of the recurrence relation,

$$a_n = a_{n-1} + 2a_{n-2}$$

with $a_0 = 2$ and $a_1 = 7$?

Solution: By Theorem 18, the characteristic equation of this recurrence relation is $r^2 - r - 2 = 0$. The characteristic roots are $r_1 = 2$ and $r_2 = -1$. Hence, the solution is the sequence $\{a_n\}$ with

$$a_n = \alpha_12^n + \alpha_2(-1)^n$$

Solving for α_1 and α_2 ,

$$\begin{aligned} a_0 &= 2 = \alpha_1 + \alpha_2 \\ a_1 &= 7 = 2\alpha_1 + (-1)\alpha_2 \end{aligned}$$

Solving these two equations gives $\alpha_1 = 3$ and $\alpha_2 = -1$. Hence, the solution to the recurrence relation is the sequence $\{a_n\}$ with

$$a_n = 3 \cdot 2^n - (-1)^n$$

■

Example 48 Recall the Fibonacci recurrence:

$$f_n = f_{n-1} + f_{n-2}$$

with $f_0 = 0$ and $f_1 = 1$. Find a closed form for f_n .

Solution: The characteristic equation of $f_n = f_{n-1} + f_{n-2}$ is $r^2 - r - 1 = 0$. Its characteristic roots are $r_1 = (1 + \sqrt{5})/2$ and $r_2 = (1 - \sqrt{5})/2$. Hence,

$$f_n = \alpha_1 \left(\frac{1 + \sqrt{5}}{2} \right)^n + \alpha_2 \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

Solving for α_1 and α_2 ,

$$\begin{aligned} f_0 &= \alpha_1 + \alpha_2 = 0 \\ f_1 &= \alpha_1 \left(\frac{1 + \sqrt{5}}{2} \right) + \alpha_2 \left(\frac{1 - \sqrt{5}}{2} \right) = 1 \end{aligned}$$

Solving the two equations gives $\alpha_1 = \frac{1}{\sqrt{5}}$ and $\alpha_2 = -\frac{1}{\sqrt{5}}$. Therefore,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left(\frac{1 - \sqrt{5}}{2} \right)^n.$$

■

Theorem 19 Let c_1 and c_2 be real numbers, with $c_2 \neq 0$. Suppose that $r^2 - c_1r - c_2 = 0$ has only one distinct root, r_0 . A sequence $\{a_n\}$ is a solution to the recurrence relation $a_n = c_1a_{n-1} + c_2a_{n-2}$ if and only if $a_n = \alpha_1 r_0^n + \alpha_2 n r_0^n$, for all $n \geq 0$, where α_1 and α_2 are constants.

Example 49 What is the solution of the recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2}$$

with $a_0 = 1$ and $a_1 = 6$?

Solution: The only root of $r^2 - 6r + 9 = 0$ is $r = 3$. Hence, the solution to this recurrence relation is,

$$a_n = \alpha_1 3^n + \alpha_2 n 3^n$$

Solving for α_1 and α_2 ,

$$\begin{aligned} a_0 &= \alpha_1 = 1 \\ a_1 &= 3\alpha_1 + 3\alpha_2 = 6 \end{aligned}$$

Solving the system of equation, $\alpha_1 = 1$ and $\alpha_2 = 1$. Therefore,

$$a_n = 3^n + n 3^n$$

■

A generalization of Theorem 18 for linear homogeneous recurrence relation with constant coefficient with degree $k \geq 1$ is given as:

Theorem 20 Let $c_1, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has k distinct roots $r_1, \dots, r_k$. Then a sequence $\{a_n\}$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$$

if and only if

$$a_n = \alpha_1 r_1^n + \dots + \alpha_k r_k^n$$

for all $n \geq 0$, where $\alpha_1, \dots, \alpha_k$ are constants.

Example 50 Find the solution of the recurrence relation

$$a_n = 6a_{n-1} - 11a_{n-2} + 6a_{n-3}$$

with $a_0 = 2, a_1 = 5, a_2 = 15$.

Solution: The characteristic equation of this recurrence relation is

$$r^3 - 6r^2 + 11r - 6 = 0$$

with roots $r_1 = 1, r_2 = 2, r_3 = 3$. The solutions are of the form,

$$a_n = \alpha_1 \cdot 1^n + \alpha_2 \cdot 2^n + \alpha_3 \cdot 3^n.$$

Solving for $\alpha_1, \alpha_2, \alpha_3$, setup the system of linear equations:

$$\begin{aligned} a_0 &= \alpha_1 + \alpha_2 + \alpha_3 = 2 \\ a_1 &= \alpha_1 + 2\alpha_2 + 3\alpha_3 = 5 \\ a_2 &= \alpha_1 + 4\alpha_2 + 9\alpha_3 = 15 \end{aligned}$$

Solving the system, $\alpha_1 = 1, \alpha_2 = -1$ and $\alpha_3 = 2$. Therefore,

$$a_n = 1 - 2^n + 2 \cdot 3^n$$

■

For the solution of a linear homogeneous recurrence relation with constant coefficient with degree $k \geq 2$, where its characteristic equation may have roots with multiplicity, it is given by:

Theorem 21 Let $c_1, \dots, c_k$ be real numbers. Suppose that the characteristic equation

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has t distinct roots $r_1, \dots, r_t$ with multiplicities $m_1, \dots, m_t$, respectively, so that for each $1 \leq i \leq t$, $m_i \geq 1$ and $m_1 + \dots + m_t = k$. Then a sequence $\{a_n\}$ is a solution to the recurrence relation

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n &= (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})r_1^n \\ &\quad + (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})r_2^n \\ &\quad + \dots + (\alpha_{t,0} + \alpha_{t,1}n + \dots + \alpha_{t,m_t-1}n^{m_t-1})r_t^n \end{aligned}$$

for all $n \geq 0$, where $\alpha_{i,j}$ are constants for $1 \leq i \leq t$ and $0 \leq j \leq m_i - 1$.

9.2 Solving Linear Nonhomogeneous Recurrence Relation with Constant Coefficients

Definition 8 A linear nonhomogeneous recurrence relation of degree k with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where $c_1, \dots, c_k$ are real numbers and $F(n)$ is a non-zero function. The recurrence relation

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

is called the associated homogeneous recurrence relation.

Example 51 The following are examples of non-homogeneous recurrence relations:

1. $a_n = a_{n-1} + 2^n$. The associated homogeneous recurrence relation is $a_n = a_{n-1}$.
2. $a_n = a_{n-1} + a_{n-2} + n^2 + n + 1$. The associated homogeneous recurrence relation is $a_n = a_{n-1} + a_{n-2}$.

Theorem 22 If $\{a_n^{(p)}\}$ is a particular solution of the nonhomogeneous linear recurrence relation with constant coefficient

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n),$$

then every solution is of the form $\{a_n^{(p)} + a_n^{(h)}\}$, where $\{a_n^{(h)}\}$ is a solution of the associated homogeneous recurrence relation

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$$

Example 52 Find all solutions of the recurrence relation $a_n = 3a_{n-1} + 2n$. What is the solution with $a_1 = 3$?

Solution: To solve this nonhomogeneous linear recurrence, need to solve its associated homogeneous linear recurrence and find a particular solution for the nonhomogeneous equation. Solving $a_n = 3a_{n-1}$, its solutions are $a_n^{(h)} = \alpha 3^n$, where α is a constant.

Now to find the particular solution. Since $F(n) = 2n$ is a polynomial in n of degree one, a trial solution is a linear function in n of the form $a_n^{(p)} = cn + d$, where c and d are constants. The equation $a_n = 3a_{n-1} + 2n$ becomes

$$cn + d = 3(c(n-1) + d) + 2n.$$

Simplifying and combining like terms gives $(2 + 2c)n + (2d - 3c) = 0$. It follows that $cn + d$ is a solution if and only if $2 + 2c = 0$ and $2d - 3c = 0$. Solving for c and d gives $c = -1$ and $d = -\frac{3}{2}$. Hence, $a_n^{(p)} = -n - \frac{3}{2}$ is a particular solution. By Theorem 22,

$$a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha \cdot 3^n.$$

When $a_1 = 3$,

$$3 = -1 - \frac{3}{2} + 3\alpha,$$

and solving α gives $\alpha = \frac{11}{6}$. Hence the solution is

$$a_n = -n - \frac{3}{2} + \frac{11}{6} \cdot 3^n$$

■

Example 53 Find all solutions of the recurrence relation

$$a_n = 5a_{n-1} - 6a_{n-2} + 7^n$$

Solution: The solutions of the associated homogeneous recurrence

$$a_n = 5a_{n-1} - 6a_{n-2}$$

are $a_n^{(h)} = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n$, where α_1 and α_2 are constants. Since $F(n) = 7^n$, a trial solution is $a_n^{(p)} = C \cdot 7^n$, where C is a constant. Substituting the terms of this sequence into the recurrence relation gives

$$C \cdot 7^n = 5C \cdot 7^{n-1} - 6C \cdot 7^{n-2} + 7^n$$

Factoring out 7^{n-2} , this gives $49C = 35C - 6C + 49$, implying that $20C = 49$, or $C = \frac{49}{20}$. Hence, $a_n^{(p)} = \frac{49}{20} \cdot 7^n$ is a particular solution. Therefore, all solutions are of the form

$$a_n = \alpha_1 \cdot 3^n + \alpha_2 \cdot 2^n + \frac{49}{20} \cdot 7^n$$

■

Theorem 23 Suppose that $\{a_n\}$ satisfies the linear nonhomogeneous recurrence relation

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} + F(n)$$

where $c_1, \dots, c_k$ are real numbers, and

$$F(n) = (b_t n^t + \cdots + b_1 n + b_0) s^n,$$

where $b_0, \dots, b_t$ and s are real numbers. When s is not a root of the characteristic equation of the associated linear homogeneous recurrence relation, there is a particular solution of the form

$$(p_t n^t + \cdots + p_1 n + p_0) s^n$$

When s is a root of the characteristic equation with multiplicity m , there is a particular solution of the form

$$n^m (p_t n^t + \cdots + p_1 n + p_0) s^n$$

Example 54 What form does a particular solution of the linear nonhomogeneous recurrence relation

$$a_n = 6a_{n-1} - 9a_{n-2} + F(n)$$

where $F(n)$ is:

1. $F(n) = 3^n$?
2. $F(n) = n^2 2^n$?
3. $F(n) = n 3^n$?
4. $F(n) = (n^2 + 1) 3^n$?

Solution: The associated linear homogeneous recurrence is $a_n = 6a_{n-1} - 9a_{n-2}$, with characteristic equation $r^2 - 6r + 9 = 0$, has a single root 3 with multiplicity 2. Since $r = 3$, then $s = 2$ is not a root. So the forms for the different $F(n)$ are

1. for $F(n) = 3^n$, $a_n^{(p)} = p_0 n^2 3^n$
2. for $F(n) = n^2 2^n$, $a_n^{(p)} = (p_2 n^2 + p_1 n + p_0) 2^n$
3. for $F(n) = n 3^n$, $a_n^{(p)} = n^2 (p_1 n + p_0) 3^n$
4. for $F(n) = (n^2 + 1) 3^n$, $a_n^{(p)} = n^2 (p_2 n^2 + p_1 n + p_0) 3^n$

■

10 Generating Functions

Definition 9 The generating function for the sequence $\{a_k\}_{k \geq 0}$ of real numbers is the infinite series

$$G(x) = a_0 + a_1x + \cdots + a_kx^k + \cdots = \sum_{k=0}^{\infty} a_kx^k.$$

Example 55 What are the generating functions for the following sequences:

1. $a_k = 3$

$$\sum_{k=0}^{\infty} 3x^k$$

2. $a_k = k + 1$

$$\sum_{k=0}^{\infty} (k+1)x^k$$

3. $a_k = 2^k$

$$\sum_{k=0}^{\infty} 2^k x^k$$

4. $\{1, 1, 1, 1, 1, 1\}$

$$1 + x + x^2 + x^3 + x^4 + x^5 = \frac{x^6 - 1}{x - 1}, \text{ when } x \neq 1$$

Note that x are just place holders for the terms of the sequence and $G(1)$ being undefined is not critical.

5. $a_k = \binom{m}{k}; 0 \leq k \leq m \text{ and } m \in \mathbb{Z}^+$

$$\binom{m}{0} + \binom{m}{1}x + \binom{m}{2}x^2 + \cdots + \binom{m}{m}x^m = \sum_{k=0}^m \binom{m}{k}x^k = (1+x)^m$$

Please see p.542 of Discrete Mathematics and Its Applications, 7th edition by Kenneth Rosen, for some useful generating functions.

10.1 Solving Counting Problems with Generating Functions

Example 56 Find the number of solutions to

$$e_1 + e_2 + e_3 = 17$$

where for each i , $1 \leq i \leq 3$, e_i is a nonnegative integer with $2 \leq e_1 \leq 5$, $3 \leq e_2 \leq 6$, and $4 \leq e_3 \leq 7$. The number of solutions indicated with the constraints is the coefficient of x^{17} in the expansion of

$$(x^2 + x^3 + x^4 + x^5)(x^3 + x^4 + x^5 + x^6)(x^4 + x^5 + x^6 + x^7)$$

Obtain a term equal to x^{17} in the product by taking a term in the first sum x^{e_1} , a term in the second sum x^{e_2} , and a term in the third sum x^{e_3} , where the exponents must add up to 17 given the constraints.

Solution: Verify that the product expansion of the three polynomials has 3 as coefficient for x^{17} . Hence, there are 3 solutions to

$$e_1 + e_2 + e_3 = 17$$

given the constraints. ■

Example 57 *In how many different ways can eight identical cookies be distributed among three distinct children if each child receives at least two cookies and no more than four cookies?*

Solution: Because each child receives at least two but no more than four cookies, for each child there is a factor equal to $(x^2 + x^3 + x^4)$ in the generating function for the sequence $\{c_n\}$, where c_n is the number of ways to distribute n cookies. Because there are three children, this generating function is

$$(x^2 + x^3 + x^4)^3$$

We need the coefficient of x^8 in this product. Using a polynomial solver, verify that the expansion has $6x^8$.

Alternatively, we can solve this analytically. Given $(x^2 + x^3 + x^4)$, we can factor out x^2 giving us $(1 + x + x^2)$. Since there are three children, we have $(1 + x + x^2)^3$. We need to find the coefficient of x^2 (since we already factored out x^6 (i.e. distributed 6 cookies)). The generating function of $(1 + x + x^2)^3$ is

$$\left(\frac{1 - x^3}{1 - x}\right)^3 = \frac{1 - 3x^3 + 3x^6 - x^9}{(1 - x)^3}$$

To find the coefficient of x^2 , look at the terms in the numerator. Every term with degree less than or equal to 2 will contribute to the coefficient of x^2 . In this case, there is only one ($1x^0 = 1$). The coefficient of the term x^k in $\frac{1}{(1 - x)^n}$ is $\binom{k + n - 1}{n - 1}$. So the coefficient of x^2 is $\binom{2 + 3 - 1}{3 - 1} = \binom{4}{2} = 6$.

Therefore, there are 6 ways to distribute eight identical cookies to the three children. ■

Looking back at

$$\left(\frac{1 - x^3}{1 - x}\right)^3 = \frac{1 - 3x^3 + 3x^6 - x^9}{(1 - x)^3}$$

if we wanted to find the coefficient of, for example x^5 , there would be 2 terms that contribute to the coefficient, $1 - 3x^3$. The coefficient would be $\binom{5 + 2}{2} - 3\binom{2 + 2}{2}$. Notice how in the second term that we subtracted 3 from $k = 5$ since the contributing term is x^3 . Solving the binomials, the coefficient of x^5 is 3.

Example 58 *Use generating functions to determine the number of ways to insert tokens worth P1, P5, and P10 into a vending machine to pay for an item that costs r pesos in both the cases when the order in which the tokens are inserted does not matter and when the order does matter.*

Solution: Consider the case when the order in which the tokens are inserted does not matter. Here, all we care about is the number of each token used to produce a total of r pesos. Because we can use any number of P1 tokens, any number of P5 tokens, and any number of P10 tokens, the answer is the coefficient of x^r in the generating function

$$(1 + x + x^2 + x^3 + \dots)(1 + x^5 + x^{10} + x^{15} + \dots)(1 + x^{10} + x^{20} + x^{30} + \dots).$$

For example, the number of ways to pay for an item costing $P75$ using $P1$, $P5$, and $P10$ tokens is given by the coefficient of x^{75} in this expansion.

When the order in which the tokens are inserted matters, the number of ways to insert exactly n tokens to produce a total of r pesos is the coefficient of x^r in $(x + x^5 + x^{10})^n$ because each of the r tokens may be a $P1$ token, a $P5$ token, or a $P10$ token. Because any number of tokens may be inserted, the number of ways to produce r pesos using $P1$, $P5$, or $P10$ tokens, when the order in which the tokens are inserted matters, is the coefficient of x^r in

$$1 + (x + x^5 + x^{10}) + (x + x^5 + x^{10})^2 + \cdots = \frac{1}{1 - (x + x^5 + x^{10})}$$

■

10.2 Solving Recurrence Relations with Generating Functions

Example 59 Solve the recurrence relation $a_k = 3a_{k-1}$ for $k = 1, 2, 3, \dots$ and initial condition $a_0 = 2$.

Solution: Let $G(x)$ be the generating function for the sequence $\{a_k\}$. Note that,

$$xG(x) = \sum_{k=0}^{\infty} a_k x^{k+1} = \sum_{k=1}^{\infty} a_{k-1} x^k$$

Using the recurrence relation,

$$\begin{aligned} G(x) &= a_0 + a_1x + a_2x^2 + \cdots \\ &= a_0 + \sum_{k=1}^{\infty} a_k x^k \\ &= a_0 + \sum_{k=1}^{\infty} 3a_{k-1} x^k \\ &= 2 + 3xG(x) \end{aligned}$$

because $a_0 = 2$ and $a_k = 3a_{k-1}$. Thus,

$$G(x) - 3xG(x) = (1 - 3x)G(x) = 2.$$

Solving for $G(x)$ show that

$$G(x) = \frac{2}{1 - 3x}.$$

Using the identity,

$$\frac{1}{1 - ax} = \sum_{k=0}^{\infty} a^k x^k,$$

we have that,

$$G(x) = 2 \sum_{k=0}^{\infty} 3^k x^k = \sum_{k=0}^{\infty} 2 \cdot 3^k x^k$$

Hence, $a_k = 2 \cdot 3^k$.

■

Example 60 Suppose that a valid codeword is an n -digit number in decimal notation containing an even number of 0s. Let a_n denote the number of valid codewords of length n . The recurrence relation was shown previously as,

$$a_n = 8a_{n-1} + 10^{n-1}$$

with initial condition $a_1 = 9$. Use generating functions to find an explicit formula for a_n .

Solution: Extend the initial condition to have $a_0 = 1$ (the empty word is valid since there is an even number of 0s). Let $G(x)$ be the generating function of $\{a_n\}_{n \geq 0}$. Starting from $n = 1$, we have

$$\begin{aligned}
 G(x) - 1 &= \sum_{n=1}^{\infty} a_n x^n = \sum_{n=1}^{\infty} (8a_{n-1} + 10^{n-1}) x^n \\
 &= \sum_{n=1}^{\infty} 8a_{n-1} x^n + \sum_{n=1}^{\infty} 10^{n-1} x^n \\
 &= 8x \sum_{n=1}^{\infty} a_{n-1} x^{n-1} + x \sum_{n=1}^{\infty} 10^{n-1} x^{n-1} \\
 &= 8x \sum_{n=0}^{\infty} a_n x^n + x \sum_{n=0}^{\infty} 10^n x^n \\
 &= 8xG(x) + \frac{x}{1-10x}
 \end{aligned}$$

Hence,

$$G(x) - 1 = 8xG(x) + \frac{x}{1-10x}$$

Solving for $G(x)$,

$$G(x) = \frac{1-9x}{(1-8x)(1-10x)}$$

By partial fractions,

$$G(x) = \frac{1}{2} \left(\frac{1}{1-8x} + \frac{1}{1-10x} \right).$$

Again, applying the identity $\frac{1}{1-ax} = \sum_{k=0}^{\infty} a^k x^k$, gives,

$$\begin{aligned}
 G(x) &= \frac{1}{2} \left(\sum_{n=0}^{\infty} 8^n x^n + \sum_{n=0}^{\infty} 10^n x^n \right) \\
 &= \sum_{n=0}^{\infty} \frac{1}{2} (8^n + 10^n) x^n
 \end{aligned}$$

Therefore,

$$a_n = \frac{1}{2} (8^n + 10^n)$$

■

11 Exercises

1. How many three digit numbers are divisible by 7?
2. How many numbers in the range $1000 \sim 9999$ do not have any repeated digits?
3. Consider the alphanumeric passwords for online accounts. Some website measure password strength when you make them. Discuss why a password is stronger:
 - (a) If it is longer (the password has more symbols)?
 - (b) If it has a mix of uppercase and lowercase letters and digits (assuming case-sensitivity)?

4. How many ways to get an odd sum from rolling two distinguishable 8-sided dice? How many if the two dice are indistinguishable?
5. Use the Inclusion-Exclusion principle to prove that if $n = pq$, where p and q are distinct prime numbers, then $\varphi(n) = (p-1)(q-1)$.
6. How many bit strings of length eight either start with a 1 or ends with 00?
7. How many bit strings of length six contain two consecutive 0's (00)? How many bit strings of length six do not contain two consecutive 0's?
8. Prove the inclusion-exclusion principle for two sets A and B by showing that $A \cup B$ can be partitioned into $A \cap B$, $A - (A \cap B)$, and $B - (A \cap B)$, and then use the sum rule for disjoint sets.
9. Prove the inclusion-exclusion principle for three sets.
10. Let n be a positive integer. If $n+1$ are chosen (without repetition) from the range 1 to $2n$, must at least one of those chosen integers be even?
11. A certain college class has 40 students. All students in the class are known to be from 17 to 34 years of age. You want to make a bet that the class contains at least x students of the same age. How large can x be made and yet be sure to win the bet?
12. Let A be a set of six distinct positive integers each of which is less than 15. Show that there must be two non-empty distinct subsets of A whose elements when added up give the same sum.
13. A student council consists of 15 students.
 - (a) In how many ways can a six-member committee be formed from the members of the council?
 - (b) Two council members always insists on serving committees together. If they cannot work together, they would not serve at all. How many ways to form a six-member committee from the members of the council with this restriction?
 - (c) Suppose the council consists of three freshmen, four sophomores, three juniors, and five seniors. How many committees of eight contain one representative from each class?
14. Let n be a positive integer. How many triple, (i, j, k) can be formed such that $1 \leq i \leq j \leq k \leq n$?
15. How many solutions are there to the inequality,

$$x_1 + x_2 + x_3 \leq 11$$
 where x_1 , x_2 , and x_3 are nonnegative integers?
16. How many strings of length 10 can be formed from the alphabet $\{0, 1, 2\}$ such that:
 - (a) The string contains exactly two 0's, three 1's, and five 2's?
 - (b) The string does not contain exactly two 0's, three 1's, and five 2's?
17. In a Five-Card Draw poker, each player is given five cards from the 54-Card French Deck (including two jokers) at the beginning of a round. How many ways can the cards be distributed at the beginning of the round if there are 5 players?
18. How many ways can n books be placed on k distinguishable shelves:
 - (a) if the books are indistinguishable copies of the same title?

(b) if no two books are the same, and the positions of the books on the shelves matter?

19. Let n be a non-negative integer. Consider the equation:

$$\sum_{i=0}^n \binom{n}{i} = 2^n$$

(a) Use mathematical induction to prove the equation. (Hint: Use Pascal's Formula in the inductive step)

(b) Use combinatorial proof to prove the equation.

20. Express $\sum_{k=0}^n \binom{n}{k} 9^k$ as x^y , $x, y \in \mathbb{Z}$.

21. The Vandermonde's Identity is given as

$$\binom{m+n}{r} = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

where $m, n \geq 1$, and $r \leq m, n$. Prove the following using Vandermonde's Identity:

$$\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2$$

22. Prove that $\forall n \geq 0, \sum_{i=0}^n \binom{n}{i} 2^{2i} = 5^n$

23. Prove that $\forall n \geq 0, \sum_{i=0}^n \binom{n}{i}^2 = \binom{2n}{n}$

24. Consider the pseudocode:

```

k ← 0
for i1 ← 1 to n do
    for i2 ← 1 to i1 do
        ⋮
        for im ← 1 to im-1 do
            k ← k + 1

```

What is the value of k after executing the pseudocode?

25. How many permutations of the 26 letters of the English alphabet contains either the string *binary* or the string *algorithm* or the string *compute*?

26. Consider the set $\{1, 2, 3, 4, 5, 6\}$

(a) Find the next 4-combination after $\{1, 2, 5, 6\}$: $\{1, 3, 4, 5\}$

(b) Find the next 4-combination after $\{1, 4, 5, 6\}$: $\{2, 3, 4, 5\}$

(c) Find the next 5-combination after $\{1, 2, 3, 4, 6\}$: $\{1, 2, 3, 5, 6\}$

(d) Find the next 3-combination after $\{2, 5, 6\}$: $\{3, 4, 5\}$